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NITI Aayog



SECTOR AT A GLANCE 07

Water Sharing and Distribution in India

Water in India

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The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Fresh water is vital but scarce in India as the country houses 18% of the world's population but only 4% of the world's water resources.¹ However, the supply and demand for water within India are also unevenly spread and effective water sharing and distribution can lead to improved food security, better public health, and greater economic gains.

Policies and acts have been implemented to ensure the fair allocation of water resources among states aided by the presence of several water sources that get shared across states, including surface water from rivers such as Cauvery, Krishna, Mahanadi, and the like.²

However, water sharing is a complex domain, and plans can face certain challenges linked to coordination, data access, and enforcement of tribunals on the one hand and ensuring ecological sustainability on the other.

Given this background, effective sharing and distribution of surface and groundwater, in a manner that is environmentally sustainable, as well as sharing of relevant data on water sources, amongst others, have been major policy focus areas. Several government initiatives, including but not limited to the National Water Policy, Interstate River Disputes Act, and the River Boards Act focus on effective water sharing and distribution.

These efforts, along with several state- and local-level policies and initiatives, seek to ensure that India adopts an equitable water-sharing approach for the betterment of the population across States.

This note describes key trends, challenges, and opportunities, as well as some thoughts on the way forward.

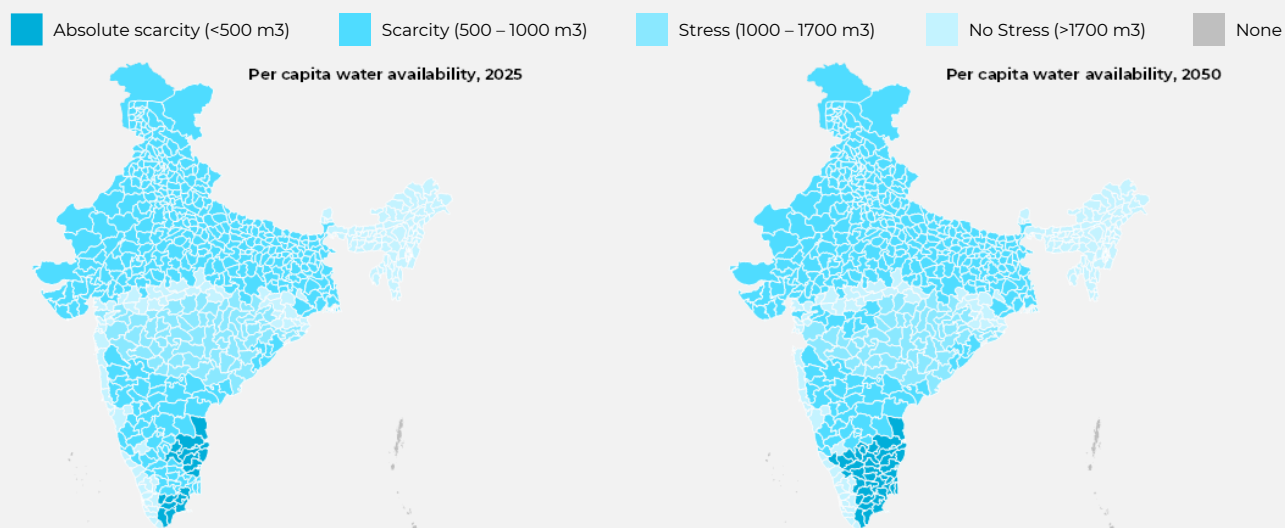
Figure 1: Key water sharing and distribution statistics



Source: World Bank, How is India addressing its water needs?, 2023; Press Information Bureau, Water insecurity, 2022; Ministry of Health and Family Welfare, WHO report highlights significant impact of 'Har Ghar Jal' Program on Public Health and Economic Savings, 2023; Ministry of Jal Shakti, Rajya Sabha Unstarred Question No. 2801, 2022

¹ World Bank, How is India addressing its water needs?, 2023
² Ministry of Jal Shakti, Acts/Tribunals, accessed December 2023

Figure 2: Per-capita water availability across regions (2025 and 2050)



Source: NITI Aayog, India Climate and Energy Dashboard, accessed December 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by ensuring effective water sharing and distribution. These revolve primarily around improved food security, better public health and economic gains.



Improved food security:

Irrigation accounts for over 80% of water use in India, and increasing agricultural productivity is critical for food security, with the population projected to reach 1.64 billion by 2050.³ Effective water transfer programs can improve agriculture production in states where irrigation water becomes a bottleneck.



Better public health:

Effective water sharing can ensure adequate water flows into water supply systems across states thereby ensuring all communities receive adequate amounts of clean water for drinking and sanitation and reducing the chance of disease from poor quality or limited water supply.



Economic gains:

The overarching economic benefits of adequate water sharing are substantial. Effective water sharing increases available water for irrigation, doubling agricultural productivity per unit of land compared to rainfed agriculture.⁴ Similarly, improved water sharing that drives industrial and transportation use can boost long-term economic growth and prosperity, while helping maintain biodiversity across states.

Challenges

This section highlights some of the major challenges and constraints facing water sharing and distribution. Efforts to resolve such challenges will ensure clarity in water sharing regulation, availability of data and better governance of transboundary rivers.



Inadequate data and monitoring:

Limited data on available resources, granted allocations, and use hinder the development of informed evidence-based sharing and distribution policies.⁵ States need to establish a system for regularly exchanging such and other relevant data and information. A continuous dialogue among stakeholders is essential for discussing water-sharing matters.⁶



Functioning of inter-state river water dispute tribunals:

A CAG report from 2017, revealed certain drawbacks in the functioning of Tribunals for settling water disputes. These included the absence of a strict time limit for the conclusion of adjudication by the Tribunal, the lack of a defined time limit for publishing the Report of the Tribunal, no upper age limit for the Chairman and other Members of the Tribunal, and considerable time taken to set up and initiate working of the tribunals.⁷

³ NITI Aayog, Composite Water Management Index 2.0, 2019

⁴ World Bank, Water in Agriculture, accessed 2023

⁵ TERI, Resolving water-sharing disputes: Legal approach has limits, 2018

⁶ TERI, Resolving water-sharing disputes: Legal approach has limits, 2018

⁷ Lok Sabha Secretariat, The Inter-State River Water Disputes (Amendment) Bill, 2017, 2017



Complexities in governing transboundary rivers:

As per a report published by the National Institute of Disaster Management, the governance and consequent use of transboundary river water in South Asia, including in India, poses complexity given concerns related to national security, sovereignty, and territoriality.⁸ Moreover, decisions related to transboundary waters tend to lose the perspective of communities whose lives and livelihoods depend on such rivers.⁹

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⁸ NIDM, Transboundary Cooperation For Water And Related Issues In South Asia, 2021
⁹ NIDM, Transboundary Cooperation For Water And Related Issues In South Asia, 2021

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee India's water-sharing and distribution efforts.



Department of Water Resources, River Development, and Ganga Rejuvenation, Ministry of Jal Shakti

The Ministry, through the implementation of the National Water Mission, aims to conserve water, minimize wastage and ensure its more equitable distribution both across and within States through integrated water resources development and management. It also houses various tribunals to navigate through current inter-state river water disputes.¹⁰

This includes some noteworthy bodies related to water sharing and distribution, and these are:

- **Tribunals:** As under the Inter-State River Water Disputes (ISRWD) Act, 1956 which is concerned with for adjudication of disputes relating to waters of inter-state rivers, the Central Government constitutes a Water Disputes Tribunal for adjudication of the said disputes which are not settled through negotiations.¹¹
- **Central Water Commission:** This is a technical organisation in the field of water resources responsible for initiating and coordinating schemes for control, conservation and utilization of these resources throughout the country, in consultation with State governments. This includes themes like flood control, irrigation, navigation, drinking water supply and water power development.¹²

Major policies and programs

This section highlights existing policies related to the blue economy and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



National Water Policy, 2012¹³

This policy promotes participatory water management, efficiency, and sustainability.¹⁴ It consists of specific provisions for sharing and management of water resources between states.



Interstate River Disputes Act, 2019¹⁵

This act provides for adjudication of disputes related to waters of inter-state rivers and river valleys by establishing tribunals to resolve such conflicts between states.



River Boards Act, 1956¹⁶

This act provides for the establishment of River Boards for regulating and developing inter-state rivers and river valleys.

¹⁰ Ministry of Jal Shakti, Implementation of National Water Mission, 2022

¹¹ Ministry of Jal Shakti, Inter-State River Water Sharing Disputes, 2018

¹² Ministry of Jal Shakti, Central Water Commission, 2023

¹³ Ministry of Jal Shakti, GoI, National Water Policy, 2012

¹⁴ Ministry of Jal Shakti, GoI, National Water Policy, 2012

¹⁵ GoI, Interstate River Disputes Act 2019, 2019

¹⁶ GoI, River Boards Act, 1956

Way forward

Water-sharing across basins is a complex activity with multi-dimensional benefits and challenges. India has made progress towards this through river interlinking programs that have been planned and are under implementation.

Some ideas that can catalyze progress include:

 Creating equitable benefit-sharing arrangements	Investing in the creation of equitable benefit-sharing arrangements between states is more likely to drive water-sharing programs that are stable and accepted by various stakeholders.
 Investing in data	Investing in data and analytics programs linked to inter-state water sharing is likely to drive more refined benchmarks and good practices.
 Investing in climate modelling	Larger investments in climate and socio-economic modelling are likely to shape water arrangements in a manner that is future-proof and does not lead to future water scarcity in presently surplus basins.

Recommended readings

Ministry of Jal Shakti. (2022). *Implementation of National Water Mission*.

<https://jalshakti-dowr.gov.in/implementation-of-national-water-mission/>

UN and CEEW. (2018). *The Perfect Storm - Pathways to Managing India's Water Sector Sustainably*.

https://www.ceew.in/sites/default/files/CEEW-UN-The-Perfect-Storm-June18_1.pdf

Ministry of Jal Shakti. (2012). *National Water Policy*.

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